

UNIT FOUR

HEALTH RELATED COMPONENTS OF FITNESS AND PRINCIPLES OF EXERCISE PRESCRIPTION

By the end of this unit you should be able to:

- Identify the five components of health related physical fitness components
- Analyse health risks associated with excessively low body fatness
- Understand health risks associated with over fatness
- Appreciate and value the benefits of regular physical exercise to healthy living.
- Recognize the means and methods of developing the health related components of physical fitness

4.1. Health Related Components of Fitness

The overall goal of a total health related physical fitness program is to optimize the quality of life.

The specific goals of this type of fitness program are:

- To reduce the risk of disease,
- to improve total physical fitness so that, daily tasks can be completed with less effort and fatigue
- aimed at improving sport performance

A health related fitness program improves sport performance by increasing **muscular strength and endurance, improving flexibility and reducing the risk of injury.**

- Exercise scientists do not always agree on the basic components of physical fitness.
- However, most do agree that the five major components of total health-related physical fitness are: cardiorespiratory endurance, muscular strength, muscular endurance, flexibility and body composition

Meaning and Concepts of Cardiorespiratory Fitness

- Cardiorespiratory fitness: (sometimes called aerobic fitness or cardiorespiratory endurance) is considered to be a key component of health-related physical fitness.
- It is a measure of a heart's ability to pump oxygen-rich blood to the working muscles during exercise. It is also a measure of the muscle's ability to take up and use the delivered oxygen to produce the energy needed to continue exercising.

- In practical terms, cardiorespiratory endurance is the ability to perform endurance-type exercise such as (distance running, cycling, swimming, etc.).
- The individuals that have achieved a high measure of cardiorespiratory endurance are generally capable of performing 30 to 60 minutes of vigorous exercise without undue fatigue.

Meaning and Concepts of Muscle Fitness

- There are two components of muscle fitness: strength and muscular endurance.
- Muscular strength: Is the maximal ability of a muscle to generate force. It is evaluated by how much force a muscle can generate during a single maximal contraction. Practically, this means how much weight that an individual can lift during one maximal effort.

- Muscular strength is important in almost all sports. Sports such as football, basketball, and events in track and field require a high level of muscular strength.
- Even non-athletes require some degree of muscular strength to function everyday life. For example, routine tasks around the home, such as lifting bags of groceries and moving furniture, require muscular strength.
- Weight training results in an increase in the size and strength of muscles.

- Muscular Endurance: Muscular endurance is defined as the ability of a muscle to generate force over and over again. Although muscular strength and muscular endurance are related they are not the same.
- These two terms can be best distinguished by examples. An excellent example of muscular strength is a person lifting a heavy barbell during one maximal muscular effort.
- In contrast, muscular endurance is illustrated by a weightlifter performing multiple lifts or repetition of a lightweight.

- Muscle Structure: Skeletal muscle is a collection of long thin cells called fibers. These fibers are surrounded by a dense layer of connective tissue called fascia that holds the individual fibers together and separates muscle from surrounding tissues.
- Muscles are attached to bone by connective tissues known as tendons. Muscular contraction causes the tendons to pull on the bones, thereby causing movement.

- **Muscle Contraction:** Muscle contraction is regulated by signals coming from motor nerves. Motor nerves originate in the spinal cord and send nerve fibers to individual muscles throughout the body.
- The motor nerve and individual muscle fiber make contact at the neuromuscular junction (where the nerve and muscle fiber meet). Note that each motor nerve branches and then connects with numerous individual muscle fibers. The motor nerve and all of the muscle fibers it controls is called a motor unit.

- Muscle contractions are classified into two major categories: **isotonic and isometric**.
- **Isotonic** (also called dynamic) contractions: are those that result in movement of a body part.
- Most exercise or sports skills utilize isotonic contractions. For example, lifting a dumbbell involves movement of a body part and is therefore classified as an isotonic contraction.

- **An isometric** (also called static) contraction: requires the development of muscular tension but results in no movement of body parts. A classic example of an isometric contraction is, the individual exerting force against a wall of a building; the muscle is developing tension but the wall is not moving and therefore neither is the body part.
- Isometric contractions occur commonly in the postural muscles of the body during sitting or standing; for instance, they are responsible for holding the head upright.

- Note that isotonic contractions can be further subdivided into **concentric, eccentric, and isokinetic contractions**.
- **Concentric contractions** are isotonic muscle contractions that result in muscle shortening. The upward movement of the arm is an example of a concentric contraction.
- In contrast, **eccentric contractions** (Also called negative contractions) are defined as contractions in which the muscle exerts force while it lengthens.

- **Isokinetic Muscle Contractions** are concentric or eccentric contractions performed at a constant speed. That is, the speed of muscle shortening or lengthening is regulated at a fixed, controlled rate. A weight-lifting machine that controls the rate of muscle shortening generally accomplishes this.

- There are three types of skeletal muscle fibers:
 - Slow twitch,
 - fast twitch, and
 - intermediate.
- These fiber types differ in their speeds of contraction and in fatigue resistance. Most human muscles contain a mixture of all three fibers types. Before beginning a strength-training program, it is helpful to have an understanding of each.

- **Slow-Twitch Fibers:** As the name implies slow-twitch fibers contract slowly and produce small amounts of force; however, these fibers are highly resistant to fatigue.
- Slow-twitch fibers, which are red in appearance, have the capacity to produce large quantities of ATP aerobically, making them ideally suited for a low-intensity prolonged exercise like walking or slow jogging.

- **Fast-Twitch Fibers:** Fast-twitch fibers contract rapidly and generate great amounts of force but fatigue quickly.
- These fibers are white and have a low aerobic capacity, but they are well equipped to produce ATP anaerobically.
- With their ability to shorten rapidly and produce large amounts of force, fast-twitch fibers are used during activities requiring rapid or forceful movement, such as jumping, sprinting, and weight lifting.

- **Intermediate Fibers:** Are although more red in color, possess a combination of the characteristics of fast- and slow-twitch fibers.
- They contract rapidly, produce great force, and are fatigue resistant due to a well-developed aerobic capacity.
- Intermediate fibers contract more quickly and produce more force than slow-twitch fibers but contract more slowly and produce less force than fast-twitch fibers

Meaning of Flexibility

- The body is flexible. It is supposed to be flexible. You must be able to bend and reach that something you dropped on the floor.
- Flexibility is the ability to move joints freely through their full range of motion. Without routine stretching muscles and tendons shorten and become tight, this can retard the range of motion around joints and impairer flexibility.
- Research suggests that flexibility is useful in preventing some types of muscle-tendon injuries and may be useful in reducing low back pain

Stretching Techniques

- Though there are a number of stretching techniques, three kinds of stretching techniques are commonly used to increase flexibility:

- ballistic,
- static, and
- proprioceptive neuromuscular facilitation.

Ballistic stretching promotes the stretch reflex and increases the risk of injury to muscles and tendons, only the static and proprioceptive neuromuscular facilitation methods are recommended.

Static Stretching

- Static stretching is extremely effective for improving flexibility and has gained popularity over the last decade. Static stretching slowly lengthens a muscle to a point at which further movement is limited (slight discomfort is felt) and requires holding this position for a fixed period of time.
- Compared with ballistic stretching, the risk of injury associated with static stretching is minimal.

- Proprioceptive Neuromuscular Facilitation
- It is a relatively new technique for improving flexibility. Proprioceptive neuromuscular facilitation (PNF) combines stretching with alternating contracting and relaxing of muscles. There are two common types of PNF stretching: contract-relax (C-R) stretching and contract-relax/antagonist contract (CRAC) stretching.
- The purpose of contracting the antagonist muscle is to promote a reflex relaxation of the muscle to be stretched.

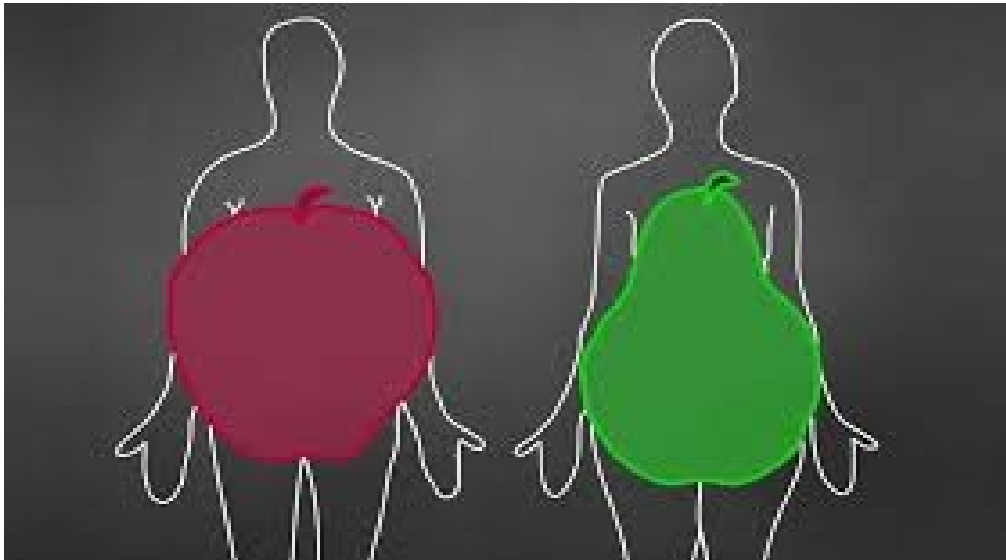
Meaning of Body Composition

- The term body composition refers to the relative amounts of fat and lean body tissue (muscle, organs, bone) found in your body.
- The rationale for including body composition as a component of health related physical fitness is that having a high percentage of body fat (a condition known as obesity) is associated with an increased risk of development of type 2 diabetes and contributes to joint stress during movement.

Health Risks Associated With Over Fatness

- Obesity has been elevated from a secondary to a primary risk factor for heart disease
- Physical fitness provides protection from the health risk of obesity
- Over fatness and obesity can contribute to degenerative disease, health problems, and even shortened life
- Heart disease is not the only disease that is associated with obesity

- Excessive abdominal fat and excessive fatness of the upper body can increase the risk of various diseases
- Apple shape/android fat / northern hemisphere for men and pear shape/southern hemisphere”/ gynoid for women



Health Risks Associated With Excessively Low Body Fatness

- Excessive desire to be thin or low in body weight can result in health problems
- Anorexia nervosa is the most severe eating disorder. If untreated, it is life threatening
- Bulimia is a common eating disorder characterized by bingeing and purging

How do people manage their body fat?

- The three most important factors to maintain a healthy body fat percentage are:
 1. Maintaining your lean muscle mass through routine strength training (minimum 2x/week for each muscle group)
 2. Maintaining a healthy well-balanced diet that is low in fat (<30% of calories from fat)
 3. Maintaining consistent sleep patterns (minimum 7 hours/night). Always seek the assistance of a medical or health and fitness professional for personal program guidance in reducing or managing your body fat.

4.2. Principles of Exercise Prescription for Health and Fitness

- Doctors often prescribe medications to treat certain diseases, and for every individual there is an appropriate dosage of medicine to cure an illness. Similarly, for each individual, there is a correct dosage of exercise to effectively promote physical fitness, called an exercise prescription.
- Exercise prescriptions should be tailored to meet the needs of the individual.
- It should include **fitness goals, mode of exercise, a warm-up, a primary conditioning period, and a cool-down**

- Establishing short-term and long-term fitness goals is an important part of an exercise prescription.
- Goals serve as motivation to start an exercise program. Further, attaining your fitness goals improves self-esteem and provides the incentive needed to make a lifetime commitment to regular exercise.
- A logical and common type of fitness goal is a performance goal. You can establish performance goals in each component of health-related physical fitness.

Establish Short-Term Goals First

Reaching short-term fitness goals is a great motivation to continue exercising. Therefore, establishment of realistic short-term goals is critical. After reaching a short-term goal, establish a new one.

Table 4.1 Fitness improvement goals

Fitness category	Current status	Short term goal	Long term goal
Cardiorespiratory fitness	Poor	Average	Excellent
Muscular strength	Poor	Average	Excellent
Muscular endurance	Very poor	Average	Good
Flexibility	Poor	Average	Good
Body composition	High fat	Moderately high	Optimal

Establish Lifetime Maintenance Goals

- In addition to short-term and long-term goals, consider establishing a fitness maintenance goal. A maintenance goal is established when your fitness goals have been met and your focus becomes remaining physically active and fit.

- Every exercise prescription includes at least one mode of exercise that is, a specific type of exercise to be performed. For example, to improve cardio respiratory fitness, you could select from a wide variety of exercise modes, such **as running, swimming, or cycling.**
- Key factors to consider when selecting an exercise mode are enjoyment, availability of the activity, and risk of injury.
- Physical activities can be classified as being either **high impact or low impact** based on the amount of stress placed on joints during the activity

4.2.3. Warm-Up

- A warm-up is a brief (5- to 15-minute) period of exercise that precedes the workout. It generally involves light calisthenics or a low-intensity form of the actual mode of exercise and often includes stretching exercises as well.
- The purpose of a warm-up is to elevate muscle temperature and increase blood flow to those muscles that will be engaged in the workout. A warm-up can also reduce the strain on the heart imposed by rapidly engaging in heavy exercise and may reduce the risk of muscle and tendon injuries.

4.2.4. Primary Conditioning Period: The Workout Plan

- The major components of the exercise prescription that make up the primary conditioning period are the mode of exercise (described earlier), frequency, intensity, and duration
- ❖ The frequency of exercise is the number of times per week that you intend to exercise.
- ❖ The intensity of exercise is the amount of physiological stress or overload placed on the body during the exercise.
- ❖ the duration of exercise, the amount of time invested in performing the primary workout.

The cool-down (sometimes called a warm-down) is a 5- to 15-minute period of low-intensity exercise that immediately follows the primary conditioning period.

- one primary purpose of a cool-down is to allow blood to be returned from the muscles back toward the heart.
- post-exercise muscle soreness may be reduced as a result of a cool-down.

4.4.1. Exercise Prescription for Cardio Respiratory Fitness

- Warm-Up
- Workout: Primary Conditioning Period

@Modes of exercise

@Frequency

@ Intensity

@ Duration

@ Intensity-Improvements in cardio respiratory fitness occur when the training intensity is approximately 50% of VO_2 max (this work rate is often called the training threshold).

Recall that training intensity can be monitored indirectly by measurement of heart rate.

The heart rate, which corresponds to exercise intensity sufficient to improve health-, related physical fitness is called the **target heart rate (THR)**. The most popular method of determining THR is the percentage of maximal heart rate (HR max) method.

This method works on the principles that exercise intensity (i.e., % VO₂ max) can be estimated by measurement of exercise heart rate.

To compute our THR using this method, simply multiply your HR max by both 90% and 70% to arrive at the high and low ends of your THR range.

For example, the maximal HR of a 20-year-old college student can be estimated by the following formula.

$$\text{HR max} = 220 - 20 = 200 \text{ beats/min}$$

The THR is then computed as $200 \text{ beats/min} \times 0.70 = 140 \text{ beats/min}$ and $200 \text{ beats/min} \times 0.90 = 180 \text{ beats/min}$

$$\text{THR} = 140 \text{ to } 180 \text{ beats/min}$$

In this example, the THR to be maintained during a workout to improve cardiorespiratory fitness is between 140 and 180 beats/min; this range of exercise intensities is sometimes called the **training sensitive zone**.

- Endurance training is a generic term that refers to any mode of exercise aimed at improving cardio respiratory fitness.
- **1. Cross Training**

Cross training is a popular form of training that uses several different training modes. It may mean running on one day, swimming on another day, and cycling on another day.

One advantage of this type of training is that cross training reduces the boredom of performing the same kind of exercise day after day.

2. Long, Slow Distance Training

Long, slow distance training, or continuous training, requires a steady, sub maximal exercise intensity (i.e., the intensity is generally around 70% HR max). It is one of the most popular cardio respiratory training techniques and can be applied to any mode of exercise.

3. Interval Training

Interval training means undertaking repeated bouts or intervals of relatively intense exercise. The duration of the intervals can be varied, but a 1- to 5-minute duration is common. Each interval is followed by a rest period, which should be equal to, or slightly greater than, the interval duration.

4. Fartlek Training

Fartlek is a Swedish word meaning, "speed play," and it refers to a popular form of training for long-distance runners. Fartlek training is much like interval training, but it is not as rigid in its work-to-rest interval ratios.

An advantage of fartlek training is that these workouts provide variety and reduce the possibility of boredom.

The Concept of Progressive Resistance Exercise (PRE) is an application of the overload principle applied to strength and endurance exercise programs. Even though the two terms can be used interchangeably.

UNIT FIVE

ASSESSMENT OF FITNESS COMPONENTS

UNIT OBJECTIVES

By the end of this unit you should be able to:

- Understand the importance of fitness assessment prior to physical exercise program**
- Identify the different means to develop own physical fitness**
- Develop skill to assess one's own physical fitness**
- Interpret the results of fitness evaluation**

INTRODUCTION

- An objective evaluation of your current fitness status is important prior to beginning an exercise-training program.
- This evaluation provides valuable information concerning your fitness strengths and weaknesses and enables you to set reasonable fitness goals. Further, testing your initial fitness level also provides a benchmark against which you can compare future evaluations.

5.1. Evaluating Health Status

Is a medical exam required before beginning a fitness program?

The answer is probably "no", for healthy college-age individuals. Although regular medical exams are encouraged for everyone, most people under 29 years of age generally do not require special medical clearance

Should individuals over 30 years old have a medical exam at the beginning of an exercise program?

The most conservative answer is "yes." This is particularly true for obese and/or sedentary individuals. The following general guidelines apply:

5.2. **Assessment of Cardiorespiratory Fitness**

- Cardio respiratory fitness is the ability to perform endurance-type exercises (e.g., running, cycling, swimming, etc.) and is considered to be a key component of health-related physical fitness.
- The most accurate means of measuring cardio respiratory fitness is the laboratory assessment of maximal oxygen consumption called VO₂ max. In simple terms, VO₂ max is a measure of the endurance capacity of both the cardio respiratory system and exercising skeletal muscles

- 5.2.1. The 1.5-Mile Run Test
- One of the simplest and most accurate means of evaluating cardio respiratory fitness is the 1.5-mile run test.
- This test was popularized by Dr. Kenneth Cooper and works on the physiological principle that people with a high level of cardio respiratory fitness can run 1.5 miles in less time than less fit individuals. The 1.5-mile run test is excellent for physically active college-age individuals.

Table 5.1 Fitness Categories for Cooper's 1.5-Mile Run Test to Determine Cardio respiratory Fitness

Fitness category	Age category					
	13-19	20-29	30-39	40-49	50-59	60+
Men						
V. poor	>15:30	>16:00	>16:30	>17:30	>19:00	>20:00
Poor	12:11-15:30	14:01-16:00	14:46-16:30	15:36-17:30	17:01-19:00	19:01-20:00
Average	10:49-12:10	12:01-14:00	12:31-14:45	13:01-13:35	14:31-17:00	16:16-19:00
Good	9:41-10:48	10:46-12:00	11:01-12:30	11:31-13:00	12:31-14:30	14:00-16:18
Excellent	8:37-9:40	9:45-10:45	10:01-11:00	10:30-11:30	11:00-12:30	11:15-13:59
Superior	<8:37	<9:45	<10:00	<10:30	<11:00	<11:15
Women						
V. Poor	>18:30	>19:00	>19:30	>20:00	>20:30	>21:00
Poor	16:55-18:30	18:31-19:00	19:01-19:30	19:31-20:30	20:01-20:30	20:31-21:31
Average	14:31-16:54	15:55-18:30	16:31-19:00	17:31-19:30	19:01-20:00	19:31-20:30
Good	12:30-14:30	13:31-15:54	14:31-16:30	15:56-17:30	16:31-19:00	17:31-19:30
Excellent	11:50-12:29	12:30-13:30	13:00-14:00	13:45-15:55	14:30-16:30	16:30-17:30
Superior	<11:50	<12:30	<13:00	<13:45	<14:30	<16:30

5.2.2. The 1-Mile Walk Test

- Another field test to determine Cardio respiratory fitness is the 1-mile walk test, which is particularly useful for sedentary individuals.
- It is a weight-bearing test, however, so individuals with joint problems should not participate. The 1-mile walk test works on the same principle as the 1.5-mile run test. That is, individuals with high levels of Cardio respiratory fitness will complete a 1-mile walk in a shorter time than those who are less conditioned

Table 5.2 Fitness Classifications for 1-Mile Walk Test

Fitness category	Age category			
	13-19	20-29	30-39	40+
Men				
Very Poor	>17:30	>18:00	>19:00	> 21:30
Poor	16:01-17:30	16:31-18:00	17:31-19:00	18:31-21:36"
Average	14:01-16:00	14:31-16:30	15:31-17:30	16:01-18:30
Good	12:30-14:00	14:31-16:30	16:31-17:30	14:00-16:00
Excellent	<12:30	<13:00	<13:30	<14:00
Women				
V. poor	>18:01	>18:31	>19:31	>20:01
Poor	16:31-18:00	17:01-18:30	18:01-19:30	19:31-20:00
Average	14:31-16:30	15:01-17:00	16:01-18:00	18:00-19:00
Good	13:31-14:30	13:31-15:00	14:01-16:00	14:31-17:59
Excellent	<13:00	<13:30	<14:00	<14:30

5.2.3. The Cycle Ergo meter Fitness Test

For those with access to a cycle ergo meter (a stationary exercise bicycle that provides pedaling resistance via friction applied to the wheel), a cycle ergo meter fitness test is an excellent means of evaluating Cardio respiratory fitness.

5.2.4 The Step Test

An alternative test to determine your cardio respiratory fitness level is the step test. The step test works on the principle that individuals with a high level of cardio respiratory fitness will have a lower heart rate during recovery from 3 minutes of standardized exercise (bench stepping) than less conditioned individuals

After completing the test, sit quietly in a chair or on the step bench. Find your pulse and count your heart rate for 30-second periods during the following recovery 'times:

1.5 minutes post exercise

2- 2.5 minutes post exercise

3- 3.5 minutes post exercise

Fitness Category	3-Minute step test recovery index	
	Women	Men
Superior	95-120	95-117
Excellent	121-135	118-132
Good	136-153	133-147
Average	154-174	148-165
Poor	175-204	166-192
V. poor	205-233	193-217

Table 5
Test

5.3 Assessment of Muscle Fitness

5.3.1 Assessing Muscular Strength

As discussed in unit 4, muscular strength is defined as the maximum amount of force you can produce during one contraction. Muscular strength not only is important for success in athletics, but also is useful for the average person in performing routine tasks at work or home.

Strength can be measured by the one-repetition maximum (1 RM) test, which measures the maximum amount of weight that can be lifted one time.

$$\text{Muscle strength score} = \frac{\text{1RM weight}}{\text{Body weight}} \times 100$$

$$\begin{aligned} \text{Therefore, muscle strength score} &= \frac{180 \text{ pounds}}{150 \text{ pounds}} \times 100 \\ &= 120 \end{aligned}$$

Using the table below, a muscle strength score of 120 on the bench press places a college age man in the “good” category.

Table 5.4 Fitness category of muscle strength score

Exercise	V. Poor	Poor	Average	Good	Excellent	Superior
Men						
Bench press	<50	50-99	100-110	111-130	131-149	>149
Biceps curl	<30	30-40	41-54	55-60	61-79	>79
Shoulder press	<40	41-50	51-67	68-80	81-110	>110
Leg press	<160	161-199	200-209	210-229	230-239	>239
Women						
Bench press	<40	41-69	70-74	75-80	81-99	>99
Biceps curl	<15	15-34	35-39	40-45	56-59	>59
Shoulder press	<20	20-46	47-54	55-59	60-79	>79
Leg press	<100	100-130	131-144	145-174	175-189	>189

5.3.2 Assessing Muscular Endurance

Muscular endurance is the ability of a muscle or muscle group to generate force over and over again.

A. The Push-Up Test

Table 5.5 Fitness category based on push-ups (1 min)

Men	Age: 20-29	Age: 30-39	Age: 40-49	Age: 50-59	Age: 60+
Excellent	54 or more	44 or more	39 or more	34 or more	29 or more
Good	45-54	35-44	30-39	25-34	20-29
Average	35-44	24-34	20-29	15-24	10-19
Poor	20-34	15-24	12-19	8-14	5-9
Very poor	20 or fewer	15 or fewer	12 or fewer	8 or fewer	5 or fewer
Women	Age 20-29	Age 30-39	Age 40-49	Age 50-59	Age:60+
Excellent	48 or more	39 or more	34 or more	29 or more	19 or more
Good	34-48	25-39	20-34	15-29	5-19
Average	17-33	12-24	8-19	6-14	3-4
Poor	6-16	4-11	3-7	2-5	1-2
Very poor	6 or fewer	4 or fewer	3 or fewer	2 or fewer	1. or fewer

B. The Sit-Up Test

The bent-knee sit-up test is probably the best field test available to evaluate abdominal muscle endurance.

Table 5.6 Fitness category based on sit-ups (1 min)

Age group (years)	Fitness category					
	V. poor	Poor	Average	Good	Excellent	Superior
Men						
17-29	<17	17-35	36-41	42-47	48-50	>50
30-39	<13	13-26	27-32	33-38	39-48	>48
40-49	<12	12-22	23-27	28-33	34-43	>43
50-59	<8	8-16	17-21	22-28	29-38	>38
60+	<6	6-12	13-17	18-24	25-35	>35
Women						
20-29	<14	14-18	29-32	33-35	36-47	>47
30-39	<11	11-22	23-28	29-34	35-45	>45
40-49	<9	9-18	19-23	24-30	31-40	>40
50-59	<6	6-12	13-17	18-24	25-35	>35
60+	<5	5-10	11-14	15-20	21-30	>30

5.4 Assessment of Flexibility

Flexibility is the ability to move joints freely through their full range of motion. Flexibility can decrease over time due to tightening of muscles and/or tendons. The key to maintaining flexibility is a program of regular stretching exercises.

5.4.1 Trunk Flexibility

The sit and reach test measures the ability to flex the trunk, which means stretching the lower back muscles and the muscles in the back of the thigh (hamstrings).

Table 5.7 Physical fitness norms for trunk flexion using the sit and reach test.

Sit and reach score	Fitness classification
-6 to -15	Very poor
-1 to -5	Poor
0 to +1	Average
+2 to +3	Good
+3 to +5	Excellent
+6 or above	Superior

5.4.2 Shoulder Flexibility

- As the name implies, the shoulder flexibility test evaluates shoulder range of motion (flexibility). The test is performed in the following manner.
- Table 5.8 Physical Fitness Norms for Shoulder Flexibility:

Right Hand Up Score	Left Hand Up Score	Fitness Classification
<0	<0	Very poor
0	0	Poor
1	+1	Average
+2	+2	Good
+3	+3	Excellent
+4	+4	Superior

5.5 Assessment of Body Composition

- Every person should possess at least a minimal amount of fat (percent body fat) for a good health.
- This fat is called essential fat and is necessary for temperature regulation, shock absorption and regulation of essential body nutrients, including vitamins A, D, E and K.
- The exact amount of body fat considered essential to normal body functioning has been debated.

5.5.1. The Skin Fold Test

Subcutaneous fat is measured using an instrument called a skin fold caliper. The skin fold test relies on the fact that over 50% of body fat lies just beneath the skin.

Therefore, measurement of representative samples of subcutaneous fat provides a means of estimating overall body fatness. Skin fold measurement to determine body fat is reliable but generally has a $\pm 3\%$ to 4% margin of error.

- One of the most accurate skin fold tests to estimate body fatness requires three skin fold measurements for both men and women.
- The anatomical sites to be measured in men (chest, triceps, and sub scapular skin folds) and the measurement sites for women (triceps, suprailium, and abdominal skin folds).
- Note that for standardization, all measurements should be made on the right side of the body.

Table 5.11 Body Composition fitness categories for men and women

Percent Body Fat	Body Composition Fitness Category
Men	
<10%	Low body fat
10-20%	Optimal range of body fat
21-25%	Moderately high body fat
26-31%	High body fat
>31%	Very high body fat
WOMEN	
<15%	Low body fat
15-25%	Optimal range of body fat
26-30%	Moderately high body Fat
31-35%	High body fat
>35%'	Very high body fat

5.5.2. Estimation of Body Composition: Field Techniques

Several quick and inexpensive field techniques exist to evaluate body composition and the risk of heart disease associated with over-fatness. Here we describe some of the more popular procedures currently in use.

A. Waist-to-Hip Circumference Ratio: recent evidence suggests that, the waist-to-hip circumference ratio is an excellent index for determining the risk of disease associated with high body fat.

Table 5.12 Waist –to –hip circumference ratio rating scale Classification.

Risk of disease	Men	Women
High risk	>1.0	>0.85
Moderately high risk	0.90-1.0	0.80-0.85
Optimal low risk of disease	<0.90	<0.80

B. Body Mass Index: Although many limitations exist, research has shown that the body mass index (BMI) is a useful technique for placing people into categories of normal or too much body fat.

B. The BMI is simply the ratio of the body weight (kilograms; kg) divided by the height (in meters) squared (m²): $BMI = \text{weight (kg)} / \text{height (m}^2\text{)}$

(Note: 1 kg = 2.2 pounds and 1 m = 39.25 inches.)

For example, if an individual weighs 64.5 kg and is 1.72 m tall, the BMI would be computed as follows:
 $64.5 \text{ kg} / (1.72 \text{ m})^2 = 64.5 / 2.96 = 21.8$

Table 5.13 Body Mass Index Classification of the Degree of Body Fatness.

Degree of obesity	BMI	
	Men	Women
Optimal body fat	<25	<27
Moderately high body fat	25 – 30	27 – 30
High body fat	31 – 40	31 – 40
Very high body fat	>40	>40

ENDDDDDDDD OF PHYSICAL FITNESS (SPSC 101)